Esercizio 1. (Martina Apicella) Prove that the following are equivalent

1. T is ω-categorical
2. T is ω-categorical over A (for some/any) set A.

Esercizio 2. (Beatrice Calderaro) Let C be a finite set. Prove that if $C \cap M \neq \varnothing$ for every model M containing A, then $C \cap \mathrm{acl}A \neq \varnothing$.

Esercizio 3. (Francesco Dam) Prove that the following are equivalent

1. T is ω-categorical
2. there is a countable model that is both saturated and atomic.

Esercizio 4. (Filippo Giugie) Let $\varphi(z) \in L(A)$ be a consistent formula. Prove that, if $a \in \mathrm{acl}(A, b)$ for every $b \models \varphi(z)$, then $a \in \mathrm{acl}A$. Prove the same claim with a type $p(z) \subseteq L(A)$ for $\varphi(z)$.

Esercizio 5. (Daniele Lanciotti) Let M be ω-saturated and $A \subseteq M$ finite. Suppose that every automorphism of $\mathcal{U}$ that fixes M setwise fixes also $\mathrm{acl}A$ setwise. Prove that $\mathrm{acl}A \subseteq \mathrm{acl}\varnothing$. (The interesting case is when M is countable.)

Esercizio 6. (Mattia Lo Mele & Gabriele Maiolino) Let $\mathcal{D} \subseteq \mathcal{U}^x \times \mathcal{U}^y$. For any $a \in \mathcal{U}^x$ and $b \in \mathcal{U}^y$ we define $\mathcal{D}_{(a,y)} = \{y : (a, y) \in \mathcal{D}\}$ and $\mathcal{D}_{(x,b)} = \{x : (x, b) \in \mathcal{D}\}$.

Assume $L(A)$ is countable. Let $\mathcal{D} \subseteq \mathcal{U}^x \times \mathcal{U}^y$ be such that for every $a \in \mathcal{U}^x$ and $b \in \mathcal{U}^y$ the sets $\mathcal{D}_{(a,y)}$ and $\mathcal{D}_{(x,b)}$ are definable over A. Prove that there is a model $M \supseteq A$ and a type $q(x, y) \subseteq L(A)$ such such that $q(M^x, M^y) = \mathcal{D} \cap \left(M^x \times M^y\right)$.

Nota. Se non ci sono errori, questo dovrebbe essere un possibile analogo del teorema di Namioka. Forse non è esattamente giusto, o forse vale qualcose di più forte. Forse è connesso con la nozione di quasi-definibilità, ma non vedo come.

Esercizio 7. (Andrea Morello) Let $a \in \mathcal{U} \smallsetminus \mathrm{acl}\varnothing$. Prove that $\mathcal{U}$ is isomorphic to some $\mathcal{V} \preceq \mathcal{U}$ such that $a \notin \mathcal{V}$.